

Banking Transaction Management System

SQL transaction management Implement the following MySQL transaction problems

Task 01 Create a database named week14a

Task 02 Create the following tables under week14a:

- a. Create a table account whose description is given below:

1 st column	account _number	string of characters of fixed size 5
2 nd column	balance	integer
3 rd column	original _balance	integer

- b. Create a move_funds table whose description is given below:

1 st column	from_acc	string of characters of fixed size 5
2 nd column	to_acc	string of characters of fixed size 5
3 rd column	transfer _amount	integer

- c. Create a move_funds_log table whose description is given below:

1 st column	account _number	string of characters of fixed size 5
2 nd column	move_fund_type	string of characters of fixed size 10
3 rd column	amount	integer
4 th column	timestamp	date time

- account_number is a foreign key referring to table account.
- move_fund_type takes values from the set {'deposit', 'withdraw'}

Task 03 populate data

- Populate data from file account.csv into table account
- Populate data from file trnx.csv into table move_funds

Task 04 As a root user, create the following users

- saradhi with a choosen password

- b. pbhaduri with a chosen password

Task 05 As a root user, provide the following privileges to the above two users

- a. Read, write, update and delete privileges to
 - account table
 - move_funds table
 - move_funds_log_1 table
- b. Lock tables privilege for the table account
- c. Execute privilege to the stored procedure transfer_funds_1¹
- d. Execute privilege to the stored procedure main_transfer²
- e. Execute privilege to the stored procedure transfer_funds_2³

Task 06 Perform the following sub-tasks. **All the three sub-tasks 6 (a), 6 (b) and 6(c) should be running simultaneously.**

- a. Run a MySQL client instance as root user and write a stored procedure transfer_funds_1 whose specifications are
 - i. Input
 - i. from_acc
 - ii. to_acc
 - iii. transfer_amount
 - ii. output: none
 - iii. Perform funds moving from one account to another account the specified amount. You must update the balance column of account table.

Note: Execute privilege to the stored procedure transfer_funds_1 to users saradhi and pbhaduri at this stage.

- b. Run a MySQL client instance as saradhi user
 - i. lock the account table
 - ii. Invoke transfer_funds_1() using the first record of move_funds table
 - iii. unlock the table
- c. Run a MySQL client instance as pbhaduri user
 - i. lock the account table
 - ii. Invoke transfer_funds_1() using the first record of move_funds table
 - iii. unlock the table

Task 07 Create a stored procedure transfer_funds_2 with the following specification:

¹Provide this privilege after creating this stored procedure at appropriate place

²Provide this privilege after creating this stored procedure at appropriate place

³Provide this privilege after creating this stored procedure at appropriate place

- a. Input arguments
 - from_acc string of characters of fixed size 5 representing from account number
 - to_acc string of characters of fixed size 5 representing to account number
 - transfer amount integer representing money being transferred
- b. Return output: none
- c. Perform the following steps:
 - i. Start a transaction
 - ii. Verify that the from_acc has a balance of at least 100. If not, rollback
 - iii. Deduct the transfer amount from balance of from_acc
 - iv. Add transfer amount to balance of to_acc.
 - v. Insert a record into the move_funds_log table about the transfer.

Task 08 Create a stored procedure main_transfer 2 with the following specification

- a. Input arguments: None
- b. Output arguments: None
- c. Read one row at-a-time from table move_funds using a MySQL CURSOR. Get the input arguments.
- d. Call the stored procedure transfer_funds 2 using input arguments from **task 08 (c)**

Task 09 Perform the following sub-tasks. These two sub-tasks 8 (a) and 8 (b) should be running simultaneously

- a. Run a MySQL client instance as saradhi user. Invoke transfer_funds 2()
- b. Run a MySQL client instance as pbhaduri user. Invoke transfer_funds 2()

Task 10 Integrity check:

- a. Compute total withdraw amount on each account number. Name this column as total withdraw
- b. Compute total deposit amount on each account number. Name this column as total deposit
- c. Original balance should be equal to balance + total deposit - total withdraw